

Stone Duality

Connecting Algebra and Topology via Logic

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Link to course website:

levinhornischer.github.io/StoneDuality_ESSLLI/

WELCOME!

- ▶ This course is an introduction to **Stone duality**.
- ▶ This is an exciting area of **logic** at the intersection of math, computer science, and philosophy.
- ▶ **Stone's theorem** says that certain algebras (Boolean algebras) are in a precise sense equivalent to certain topological spaces (now called Stone spaces).
- ▶ Behind this mathematical fact lies a deep conceptual insight: that the algebraic and the spatial perspective are just **two sides of the same coin**!
- ▶ Understanding this is the goal of the course.

- ▶ **Schedule:** in the first week of ESSLLI 2026 (August 3–7, 2026), from 09:00 to 10:30 in room C206.
- ▶ **About me:** I'm Levin Hornischer (he/him), an assistant professor at the Munich Center for Mathematical Philosophy (MCMP) at LMU Munich. Before that, I was at the Institute for Logic, Language and Computation (ILLC) at the University of Amsterdam, where I also did my PhD.
- ▶ **Comments and feedback** are most welcome: just email me at Levin.Hornischer@lmu.de.
- ▶ Please interrupt for **questions any time!**

- There is a website for the course:

levinhornischer.github.io/StoneDuality_ESSLI/



- These slides are intended as **handouts** for the lecture (hence more cluttered than presentation slides—sorry!).
- I have given a full semester course on Stone duality, on which this course is based. You find the **lecture notes** linked on the course website.
- **Prerequisites:** An introductory course in logic (specifically classical propositional logic) and some familiarity with mathematics (ideally, but not necessarily, having seen elementary concepts of topology and algebra).

- ▶ A wonderful recent textbook on Stone duality is [GG24] and further great textbooks on—or closely related to—Stone duality include [BD75; DP02; Vic89; GH08; Giv14; Grä11; Grä03; AJ94].
- ▶ Research monographs on duality theory and related topics include [Joh82; Gie+03; DST19; Gou13; PP12; ORR15].
- ▶ Extensions of Stone duality from propositional to first-order logic include [Mak87; Lur20; AF13].
- ▶ Stone duality also is an excellent pathway into category theory [Lei14; Smi25], and in particular categorical logic [Pit01] and topos theory [Vic07].

PLAN

1. Introduction and motivation: the idea of a duality
2. Boolean algebras
3. Stone spaces
4. Stone's theorem and proof idea
5. Applications and generalizations

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

THE GENERAL IDEA OF A DUALITY

- ▶ The general idea of a duality is that there are two, in a sense complementary perspectives on the same underlying phenomenon.
- ▶ Metaphorically speaking, there are **two sides** (the perspectives) of the **same coin** (the phenomenon).
- ▶ This occurs throughout the sciences. We consider four brief examples. (Also see [Geh09; Abr91; Vic89].)

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

- ▶ Now we all know it, but it is remarkable: there is a close correspondence between [algebra](#) and [geometry](#).
- ▶ Algebraic objects, like the polynomial $x^2 + y^2 = 1$, correspond to spatial objects, in this case to the unit circle.
- ▶ We can solve geometric problems, e.g., which ratios and angles can be constructed with a compass and straightedge, by [translating](#) them to the algebraic side and then using algebraic methods (here Galois theory).
- ▶ The duality of algebra and geometry is at the heart of algebraic geometry and bears quite some similarities to Stone duality [[Joh82](#); [MM92](#); [Vic07](#)].

- ▶ On the one hand, a physical system is described by its **state space**, i.e., the collection of states that the system could be in.
- ▶ On the other hand, the system is also described by the **observations** we can make about it.
- ▶ We would hope that these two perspectives on the system are ‘dual’ to each other:
 - ▶ The states should be **determined** by the observations that they give rise to; and
 - ▶ the observations should be **determined** by the states that give rise to them [e.g. **Str08**, p. 24].
- ▶ The formal duality (e.g., Gelfand–Naimark) is different in detail from Stone duality but similar in spirit. (For more, see, e.g., [**Rue11**; **RBW15**; **WW23**; **DB25**].)

- ▶ Computer programs are written in a programming language. Much like for sentences in a natural language, we can ask what their meaning is.
- ▶ The meaning of a program is called its **denotation** and is, roughly, the input-output function that it computes.
- ▶ But we can also describe the program by its **program logic**:
 - ▶ e.g., given an even number as input, the program outputs a prime number.
 - ▶ More generally: **Hoare triples** ‘if currently P holds, then running the program C will result in Q being satisfied’.
- ▶ The far-reaching insight of Abramsky [Abr91] was that we again should have a duality between the denotational semantics and the program logic.

- ▶ The central question of philosophy of language is: what is the **meaning** of sentences?
- ▶ One prominent view is **possible worlds semantics**: The meaning of a sentence is the set of **possible worlds** in which the sentence is true.
- ▶ On a broadly **inferentialist** position, we describe sentence meanings by how they relate to each other: which implies which, what conjunctions are, etc. This data is recorded in the set of all sentence meanings together with the implication relation, conjunction, negation, etc.—aka the **algebra of propositions**.
- ▶ Duality talk is surprisingly scarce in philosophy, but given the success in other fields it holds much promise!

<i>Field</i>	<i>Phenomenon</i>	<i>Space</i>	<i>Algebra</i>
Mathematics	Geometry	Spatial object	Polynomial
Physics	Physical system	State space	Observations
Computer Science	Program	Denotation	Program logic
Philosophy	Meaning	Possible worlds	Propositions

- ▶ **Algebra:** Set of observations, propositions, etc. with **operations** (e.g., conjunction $\wedge$) that satisfy certain **laws** (e.g., $a \wedge b = b \wedge a$).
 - ▶ A specific type of such an algebra we will make precise as a Boolean algebra.
- ▶ **Space:** Set of **points** (e.g., states, possible worlds, etc.) with some **spatial structure**, e.g., a notion of distance between points.
 - ▶ A specific type of such a space we will make precise as a Stone space, which is a particular kind of topological space.

KEY IDEA OF DUALITY

- ▶ **Duality**: A bijective correspondence between the class of spaces and the class of algebras under consideration, i.e.,
 - ▶ a method to translate a space into an algebra, and
 - ▶ a method to translate an algebra into a space,
 - ▶ such that translating forth and back is isomorphic to where one started.
- ▶ In this sense of **intertranslatability**, the spatial perspective is equivalent to the algebraic perspective and **both describe the same phenomenon**.
- ▶ In our specific case, **Stone's theorem** makes this precise as: the category of Boolean algebras is dually equivalent to the category of Stone spaces.
- ▶ Such a duality not only provides a deep **conceptual link** between the two perspective. It also can help practically by **translating a problem** from one side to the other, where different methods and intuitions are available (cf. again geometry and algebra).

EXAMPLE: POSSIBLE WORLDS VS PROPOSITIONS I

We do not have the precise concepts yet, but let's sketch how the duality looks like in the philosophy example.

- ▶ Translation from spaces to algebras:
 - ▶ Assume a space X of possible worlds. According to possible worlds semantics, a proposition a is a set of possible worlds, i.e., $a \subseteq X$.
 - ▶ Conversely, is every subset $a \subseteq X$ a proposition? This is not commonly discussed, but we will see: only if a is “closed under [similarity](#)” (technically: a [clopen](#) subset in the topology on X).
 - ▶ Proposition a implies proposition b iff in all worlds $x \in X$ in which a is true, also b is true, i.e., $a \subseteq b$.
 - ▶ The conjunction of two propositions a and b simply is the intersection $a \cap b$. Similarly for other operations.
 - ▶ So the space X translates to the [algebra](#) $\text{Clp}(X) = (A, \subseteq, \cap, \dots)$, where A is the set of all $a \subseteq X$ that are “closed under similarity”.

EXAMPLE: POSSIBLE WORLDS VS PROPOSITIONS II

- ▶ Translation from algebras to spaces:
 - ▶ Assume an algebra $A = (A, \leq, \wedge, \dots)$ of propositions.
 - ▶ According to [Ersatzism](#), given a language/propositions, we can construct ‘substitute possible worlds’ as maximally consistent sets of propositions.
 - ▶ So an Ersatz possible world is a set $F \subseteq A$ such that
 - (1) F is closed under implication and conjunction and does not contain the logical contradiction $\perp$, and
 - (2) for any other such set $G \subseteq A$, if $F \subseteq G$, then $F = G$.(We will call such an F a [maximal filter](#).)
 - ▶ Let $\text{Spec}(A)$ be the space of all Ersatz possible worlds (we will see that there is a natural way to define a topology on $\text{Spec}(A)$).

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

EXAMPLE: POSSIBLE WORLDS VS PROPOSITIONS III

- ▶ Translating a space X of possible worlds to the algebra $\text{Clp}(X)$ and then back to Ersatz space $\text{Spec}(\text{Clp}(X))$ yields a space “isomorphic” to X .
 - ▶ We need, in particular, a bijection $f : X \rightarrow \text{Spec}(\text{Clp}(X))$.
 - ▶ We map a world x to the set $f(x) = \{a \in \text{Clp}(X) : x \in a\}$ of all (possible worlds semantics) propositions that x makes true.
- ▶ Translating an algebra A to Ersatz worlds $\text{Spec}(A)$ and then back to the algebra $\text{Clp}(\text{Spec}(A))$ yields an algebra isomorphic to A .
 - ▶ We need, in particular, a bijection $h : A \rightarrow \text{Clp}(\text{Spec}(A))$.
 - ▶ We map a proposition a to the set $h(a) = \{F \in \text{Spec}(A) : a \in F\}$ of Ersatz worlds that make true a .

Now we develop these ideas formally.

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

**Boolean
algebras**

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OVERVIEW OF THE SECTION

- ▶ We formally define Boolean algebras.
- ▶ We motivate how they describe sets of propositions or binary experiments.
- ▶ We consider three important examples: the two-element Boolean algebra, powerset Boolean algebras, and the Lindenbaum–Tarski algebra.
- ▶ Finally, we define ‘translations’ between Boolean algebras, i.e., Boolean algebra homomorphisms.

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone’s theorem
and proof idea

Applications
and
generalization

Definition

A **Boolean algebra** is a structure $(A, \vee, \wedge, \neg, \perp, \top)$ where A is a set and $\vee, \wedge : A \times A \rightarrow A$ and $\neg : A \rightarrow A$ and $\perp, \top \in A$ are such that for $a, b, c \in A$:

1. Lattice axioms:

1.1 Commutativity: $a \vee b = b \vee a$ and $a \wedge b = b \wedge a$

1.2 Associativity: $a \vee (b \vee c) = (a \vee b) \vee c$ and $a \wedge (b \wedge c) = (a \wedge b) \wedge c$

1.3 Idempotence: $a \vee a = a$ and $a \wedge a = a$

1.4 Absorption: $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$.

2. Distributivity:

2.1 $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

2.2 $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$.

3. Bounds: $a \wedge \perp = \perp$ and $a \vee \top = \top$

4. Complements: $a \wedge \neg a = \perp$ and $a \vee \neg a = \top$.

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

Definition (continued)

We often just write A for the Boolean algebra $(A, \vee, \wedge, \neg, \perp, \top)$. For $a, b \in A$, we define

$$a \leq b :\Leftrightarrow a \wedge b = a \Leftrightarrow a \vee b = b.$$

This is a partial order on A (i.e., $\leq$ is a reflexive, transitive, and antisymmetric relation on A).

(Exercise: Prove that $\leq$ indeed is a partial order. Prove the equivalence $a \wedge b = a \Leftrightarrow a \vee b = b$.)

[Welcome](#)[Introduction
and motivation](#)[Boolean
algebras](#)[Stone spaces](#)[Stone's theorem
and proof idea](#)[Applications
and
generalization](#)

► Intuition:

- One can think of A as a set of propositions, observations, binary experiments, etc.
- for which there are ‘logical’ operations $\vee, \wedge, \neg, \perp, \top$.
- E.g., given propositions (or experiments) a and b , there also is the proposition (or experiment) $a \wedge b$, which is true (or comes out positive) iff both a and b are true (or come out positive).
- The order $\leq$ describes implications between propositions (or experiments). After all, if a is equivalent to $a \wedge b$ (i.e., $a = a \wedge b$), this just says that a implies b .
- The propositions (or experiments) should behave ‘classically’ in the sense that, e.g., they commute $a \vee b = b \vee a$, are exhaustive $a \vee \neg a = \top$, etc.

- ▶ Names of the axioms:
 - ▶ The bound axioms say that $\perp$ is the $\leq$ -least element and that $\top$ is the $\leq$ -greatest element.
 - ▶ Distributivity: If we write $\wedge$ as $\times$ (multiplication) and $\vee$ as $+$ (addition) axiom 2.2 is well-known distributivity $x(y + z) = xy + xz$.
- ▶ Some axioms are superfluous:
 - ▶ e.g., distributivity is self-dual, i.e., (2.1) is equivalent to (2.2).
- ▶ More general structures:
 - ▶ A lattice is a structure $(A, \vee, \wedge)$ that satisfies the lattice axioms.
 - ▶ A bounded lattice is a structure $(A, \vee, \wedge, \perp, \top)$ that satisfies the lattice axioms and the bounds axioms.
 - ▶ A (bounded) lattice is distributive if it additionally satisfies the distributivity axiom.

- ▶ A Boolean algebra also can equivalently be described just in terms of the order: A partial order such that
 - ▶ any finite subset S has a least upper bound $\bigvee S$ and a greatest lower bound $\bigwedge S$ (so $\perp = \bigvee \emptyset$ and $a \vee b = \bigvee \{a, b\}$, similarly for $\top$ and $\wedge$),
 - ▶ it satisfies the distributivity axiom $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$, and
 - ▶ every element a has a complement, i.e., there is another element b such that $a \wedge b = \perp$ and $a \vee b = \top$.

IMPORTANT EXAMPLE 1: THE TWO-ELEMENT BOOLEAN ALGEBRA

- ▶ The simplest nontrivial Boolean algebra is the two-element Boolean algebra $\mathbf{2} = (\{0, 1\}, \vee, \wedge, \neg, 0, 1)$ with the usual Boolean truth-functions

$\vee$	0	1
0	0	1
1	1	1

$\wedge$	0	1
0	0	0
1	0	1

$\neg$	
0	1
1	0

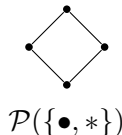
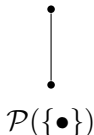
- ▶ Notation: In set theory, $0 := \emptyset$ and $n := \{0, \dots, n-1\}$. So $2 = \{0, 1\}$.
- ▶ The order $\leq$ is given by $0 \leq 1$, and we depict it as



- ▶ In fact, every Boolean algebra is a subalgebra of a product of $\mathbf{2}$. So, in a sense, $\mathbf{2}$ generates all Boolean algebras! (But we do not show this here.)

IMPORTANT EXAMPLE 2: THE POWERSET

- ▶ If X is a set, then the power set $\mathcal{P}(X) = (\mathcal{P}(X), \cup, \cap, \cdot^c, \emptyset, X)$ is a Boolean algebra with the operations of union, intersection, and set-theoretic complement (for $a \subseteq X$, $a^c = X \setminus a$).
- ▶ The order is inclusion: For $a, b \subseteq X$, $a \leq b$ iff $a \cap b = a$ iff $a \subseteq b$.
- ▶ Cf. X is a set of possible worlds and $a \subseteq X$ are propositions.
- ▶ So **2** is simply the powerset Boolean algebra of a one-element set $\{\bullet\}$.
- ▶ The powerset of the empty set $\emptyset$ has one element. The powerset of a two-element set $\{\bullet, *\}$ has four elements:



IMPORTANT EXAMPLE: THE LINDENBAUM–TARSKI ALGEBRA

- ▶ Let $\mathcal{L}$ be the set of sentence of classical propositional logic formed by

$$p \mid \varphi \vee \psi \mid \varphi \wedge \psi \mid \neg\varphi \mid \perp \mid \top$$

- ▶ Take your favorite proof system for classical propositional logic and write $\varphi \vdash \psi$ if ψ is derivable from φ .
- ▶ Define the equivalence relation $\varphi \equiv \psi$ by $\varphi \vdash \psi$ and $\psi \vdash \varphi$.
- ▶ Let $\mathcal{L}/\equiv := \{[\varphi] : \varphi \in \mathcal{L}\}$ be the quotient, with $[\varphi] := \{\psi : \varphi \equiv \psi\}$
- ▶ Define $[\varphi] \vee [\psi] := [\varphi \vee \psi]$, $[\varphi] \wedge [\psi] := [\varphi \wedge \psi]$, and $\neg[\varphi] := [\neg\varphi]$.
(Exercise: Check that this is well-defined.)
- ▶ Then $(\mathcal{L}/\equiv, \vee, \wedge, \neg, [\perp], [\top])$ is a Boolean algebra, called the Lindenbaum–Tarski algebra. (Cf. inferentialism.)

Definition

Let $A = (A, \vee_A, \wedge_A, \neg_A, \perp_A, \top_A)$ and $B = (B, \vee_B, \wedge_B, \neg_B, \perp_B, \top_B)$ be two Boolean algebras. A **Boolean algebra homomorphism** is a function $h : A \rightarrow B$ such that, for all $a, b \in A$,

1. $h(a \vee_A b) = h(a) \vee_B h(b)$
2. $h(a \wedge_A b) = h(a) \wedge_B h(b)$
3. $h(\neg_A a) = \neg_B h(a)$
4. $h(\perp_A) = \perp_B$
5. $h(\top_A) = \top_B$

If h additionally is injective (resp., bijective), we call it a Boolean algebra **embedding** (resp., **isomorphism**). Two Boolean algebras are **isomorphic** if there is an isomorphism between them.

[Welcome](#)[Introduction
and motivation](#)[Boolean
algebras](#)[Stone spaces](#)[Stone's theorem
and proof idea](#)[Applications
and
generalization](#)

- ▶ Intuition: Translation from one language/set of propositions A to another B . Preservation conditions ensure compositionality: roughly, to translate a complex proposition in A , first translate its atoms and then build it up again in B .
- ▶ Boolean algebra homomorphism are order-preserving: If $a \leq_A b$, then $h(a) \leq_B h(b)$. (Exercise: prove this.)
- ▶ Not all conditions need to be required: For example, h already is a Boolean algebra homomorphism once it preserves $\neg, \wedge, \perp$. (Exercise: prove this.)

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OVERVIEW OF THE SECTION

- ▶ In this section, we define what Stone spaces are.
- ▶ We first start by defining topological spaces (and motivate the definition)
- ▶ We define further notions: when a space is Hausdorff, zero-dimensional, and compact.
- ▶ Then we define Stone spaces as compact, zero-dimensional Hausdorff topological spaces.
- ▶ Finally, we define continuous maps between topological spaces.

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

- ▶ When we hear of ‘space’, we naturally think of the three-dimensional space we live in, i.e., the Euclidean space $\mathbb{R}^3$ whose points $x = (x_1, x_2, x_3)$ are described by the values on the x -axis, the y -axis, and the z -axis.
- ▶ But there also are other spaces:
 - ▶ The surface of a sphere: its points are those (x_1, x_2, x_3) with $x_1^2 + x_2^2 + x_3^2 = 1$. But its geometry is different: for instance, the angles of a triangle add up to more than 180 degrees.
 - ▶ The spacetime that we live in according to general relativity: its points $x = (x_1, x_2, x_3, x_4)$ are four-dimensional—with three spatial and one temporal component—and its geometry is given by a metric tensor.
 - ▶ And there are even wilder spaces, in which it might not even make sense to speak of a ‘geometry’ (e.g., angles between lines), but only of ‘spatial’ properties (e.g., continuous paths from one point to another).

- ▶ Mathematicians came up with a general definition of a topological space that includes all these examples.
- ▶ It is very abstract, but we will see that it can recover all the relevant spatial concepts.

Definition

A **topological space** is a pair (X, τ) where X is a set and τ is a collection of subsets of X such that

1. $\emptyset$ and X are in τ
2. If $U, V \in \tau$, then $U \cap V \in \tau$
3. If $U_i \in \tau$ is a collection of sets indexed by a set I , then $\bigcup_{i \in I} U_i \in \tau$.

We also call τ a topology on X .

Definition (continued)

If (X, τ) is a topological space, we call the elements of X **points**. The elements of τ are called **open sets** (or **opens**). Their complements, i.e., sets of the form $C = X \setminus U$ for $U \in \tau$, are called **closed sets**. A subset $K \subseteq X$ that is both open and closed (i.e., $K \in \tau$ and $K^c \in \tau$) is called **clopen**.

We just speak of the topological space X if τ is clear from context. Then we write $\Omega(X)$ for the opens of X . The collection of closed (resp. clopen) subsets of X is denoted $\mathcal{C}(X)$ (resp. $\text{Clp}(X)$).

[Welcome](#)[Introduction
and motivation](#)[Boolean
algebras](#)[Stone spaces](#)[Stone's theorem
and proof idea](#)[Applications
and
generalization](#)

1. The Euclidean topology on $\mathbb{R}^3$ is given by calling those sets $U \subseteq \mathbb{R}^3$ open that allow some ‘wiggle-room’, i.e.,

► for all $x \in U$, there is $\epsilon > 0$ such that for all $x' \in \mathbb{R}^3$ with $d(x, x') < \epsilon$, we have $x' \in U$.

Here the distance between two points $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ is, as usual,

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2}.$$

2. For a set X , the **discrete** topology on X is $\tau := \mathcal{P}(X)$.
3. For a set X , the **indiscrete** topology on X is $\tau := \{\emptyset, X\}$.

With just ‘open set talk’ we can recover many spatial concepts: Let (X, τ) be a topological space.

- ▶ **Interior, closure, boundary.** If $S \subseteq X$ is a subset, the interior $\text{Int}(S)$ of S is the largest open set contained in S , and the closure $\text{Cl}(S)$ of S is the smallest closed set containing S . The **boundary** ∂S of S is $\text{Cl}(S) \setminus \text{Int}(S)$.
- ▶ **Neighborhood.** A subset $S \subseteq X$ is a **neighborhood** of a point $x \in X$ if $x \in \text{Int}(S)$. An **open neighborhood** of x is an open set that is a neighborhood of x . (If clear from context, we drop ‘open’.)
- ▶ **Dense.** A subset $S \subseteq X$ is **dense** (in X) if for all points $x \in X$ and open neighborhoods U of x , there is a point $s \in S$ with $s \in U$. Equivalently, $\text{Cl}(S) = X$.

- ▶ **Convergence.** A sequence $(x_n)_{n \in \mathbb{N}}$ of points in X **converges** to a point $x \in X$ if for all open neighborhoods U of x , there is $N \geq 0$ such that, for all $n \geq N$, we have $x_n \in U$. We also say that x is the **limit** of the sequence (x_n) .
- ▶ **Generated topology.** Any collection $\mathcal{S}$ of subsets of a nonempty set Y **generates** a topology $\langle \mathcal{S} \rangle$ on Y : namely, the smallest topology on Y that contains all subsets in $\mathcal{S}$.
- ▶ **Subbase.** Given the topology τ on X , a collection $\mathcal{S}$ of subsets of X is called a **subbase** of τ if $\tau = \langle \mathcal{S} \rangle$. So the opens of τ are arbitrary unions of finite intersection of subbasic elements.

- **Base.** A **base** for the topology τ is a collection $\mathcal{S} \subseteq \tau$ such that for every point $x \in X$ and every open neighborhood U of x , there is $V \in \mathcal{S}$ such that $x \in V \subseteq U$. Equivalently, a base is a collection of open subsets of X such that every open set is a union of elements from the base.

- ▶ There is an important classification of topological spaces according to which so-called [separation axioms](#) they satisfy.
- ▶ There are many such axioms and they all are of the form that two distinct points can be—in various senses—separated by the topology.

Definition

Let (X, τ) be a topological space.

1. X is T_0 (aka [Kolmogorov](#)) if, for all $x \neq y$ in X , there is an open $U \subseteq X$ such that U contains exactly one of x and y .
2. X is T_1 (aka [Fréchet](#)) if, for all $x \neq y$ in X , there is an open $U \subseteq X$ such that $x \in U$ and $y \notin U$.
3. X is T_2 (aka [Hausdorff](#)) if, for all $x \neq y$ in X , there are disjoint opens $U, V \subseteq X$ such that $x \in U$ and $y \in V$.

Compactness formalizes the idea a space does not extend infinitely but has finite bounds.

Definition (Compactness)

Let X be a topological space. If $S \subseteq X$ is a subset, an **open cover** $\mathcal{U}$ of S is a collection of open sets such that $S \subseteq \bigcup_{U \in \mathcal{U}} U$. A subset $S \subseteq X$ is **compact** if every open cover $\mathcal{U}$ of S contains a finite subcover, i.e., there is a finite subset $\mathcal{U}_0 \subseteq \mathcal{U}$ such that $\mathcal{U}_0$ is an open cover of S . The space X is called **compact** if $S := X$ is compact.

(An equivalent definition: via **finite intersection property**.)

(ZERO-) DIMENSIONALITY

- ▶ There is a purely topological way to determine the dimension of a space. The Euclidean space $\mathbb{R}^3$ then indeed gets dimension 3.
- ▶ (In fact, there are several ways to do this: small inductive dimension, large inductive dimension, and the Lebesgue covering dimension, see [Eng89, ch. 7].)
- ▶ Here we only need what it means for a space to have dimension zero. In that special case, the definition becomes the following.

Definition

A topological space (X, τ) is **zero-dimensional** iff X has a base consisting of clopen sets.

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

Finally, we can define what a Stone space is.

Definition

A Stone space is a topological space that is compact, zero-dimensional and Hausdorff.

Examples:

- Every finite set with the discrete topology is a Stone space.

IMPORTANT EXAMPLE: CANTOR SPACE

- **Points:** The set $2^{\mathbb{N}}$ of all binary sequences (i.e., functions $\mathbb{N} \rightarrow 2 = \{0, 1\}$)
- **Topology:** The topology τ generated by the cylinder sets: given a finite binary string $\sigma = \sigma(0), \dots, \sigma(n-1)$, its cylinder set

$$[\sigma] := \{x \in 2^{\mathbb{N}} : \forall k \in \{0, \dots, n-1\}. x(k) = \sigma_k\},$$

is the set of all binary sequences that have σ as initial segment.

- ▶ Finally, we consider what it means to continuously map points of one space to another.
- ▶ Again, this has a swift abstract definition, which we then illustrate in more intuitive terms.

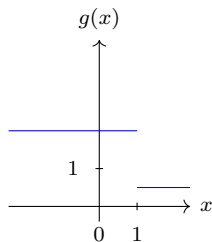
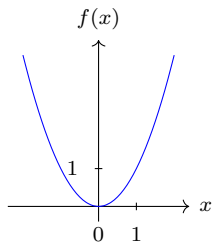
Definition

Let X and Y be topological spaces and $f : X \rightarrow Y$ a function. We say f is **continuous** if, for all open subsets V of Y , the preimage $f^{-1}(V) = \{x \in X : f(x) \in V\}$ is an open subset of X .

A function $f : X \rightarrow Y$ is a **homeomorphism** (the topologists' name for isomorphism) if f is continuous and has a continuous inverse, i.e., f is a bijection and both f and f^{-1} are continuous. Two spaces are homeomorphic if there is a homeomorphism between them.

CONTINUOUS FUNCTIONS: INTUITION

- ▶ Maybe you have encountered continuous functions before but in the more specific **epsilon–delta definition**: It says $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous if
 - ▶ For every $x \in \mathbb{R}$ and every $\epsilon > 0$, there is $\delta > 0$ such that, for all $y \in \mathbb{R}$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \epsilon$.
- ▶ This captures the idea that, to draw the graph of the function, you do not have to lift your pen.
- ▶ This ‘hands-on’ definition (pun intended) is a special case of the abstract definition.



OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

OUTLINE

Introduction and motivation

Boolean algebras

Stone spaces

Stone's theorem and proof idea

Applications and generalization

Stone Duality

Levin
Hornischer

Welcome

Introduction
and motivation

Boolean
algebras

Stone spaces

Stone's theorem
and proof idea

Applications
and
generalization

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